

Lecture 3.1: The Independence of Motion

Unit 3: The World in Two Dimensions

Today's Mission

- **Bridge Back:** In Unit 2, we mastered the language and equations for motion in one dimension (straight lines). Today, we take our first step into the real world, where things fly in arcs and curves.
- **Thinking Lens:** We will use **Systems and System Models** to understand a complex motion by breaking it into two simple, independent models.
- **Essential Questions:**
 - How can a complex, two-dimensional motion be broken down into simpler parts?
 - Do the vertical and horizontal motions of a projectile affect each other?
 - How can we use our 1D kinematic equations to solve 2D problems?

The Big Idea: The Independence of Motion

The single most important rule in projectile motion is this:

The horizontal motion of an object is completely independent of its vertical motion.

This means we can analyze a complex curved path as two separate, simultaneous, and much simpler straight-line motion problems.

The Horizontal Story: Constant Velocity

Let's consider a cannonball fired from a cliff, and let's ignore air resistance.

- **Question:** Once the cannonball leaves the barrel, are there any forces pushing or pulling it horizontally?
- **Answer:** No. Gravity only pulls downward.
- **Conclusion:** With no horizontal forces, the horizontal acceleration (a_x) is zero. This means the horizontal velocity (v_x) does not change. It is constant.
- **Equation for horizontal motion:**

- $$\Delta \vec{x} = \vec{v}_x t$$

The Vertical Story: Constant Acceleration

Now let's look at the vertical motion only.

- **Question:** Once the cannonball leaves the barrel, what is the only force acting on it?
- **Answer:** The force of gravity ($\vec{F}_g$), which pulls it straight down.

- **Conclusion:** The cannonball experiences a constant downward acceleration, $\vec{a}_g$.
- For the vertical dimension, we use our familiar kinematic equations for constant acceleration:

- ○
$$\vec{v}_{y,f} = \vec{v}_{y,i} + \vec{a}_g t$$

- ○
$$\Delta \vec{y} = \vec{v}_{y,i} t + \frac{1}{2} \vec{a}_g t^2$$

- ○
$$\vec{v}_{y,f}^2 = \vec{v}_{y,i}^2 + 2 \vec{a}_g \Delta \vec{y}$$

The Golden Link: Time

- If the horizontal and vertical motions are separate, what connects them?
- **Time (t)**
- An object can only travel horizontally for the same amount of time it is in the air.
The time it is in the air is determined by its vertical motion.
- Time is the common variable that links the two independent motions.

To solve problems, we often:

- Use the vertical information to solve for time.
- Use that time to solve for the horizontal distance.

Example: The Cliff Problem

A stone is thrown horizontally with a speed of 15.0 m/s from the top of a cliff that is 78.4 m high.

- A) How long does it take the stone to reach the bottom of the cliff?
- B) How far from the base of the cliff does the stone land?

Strategy:

- Set up a table of knowns for the x and y dimensions.
- The stone is thrown horizontally, so $v_{y,i} = 0 \text{ m/s}$.
- Use downward as negative: $\Delta y = -78.4 \text{ m}$ and $a_g = -9.80 \text{ m/s}^2$.
- Solve for time using vertical motion, then use time to find horizontal distance.

Example Solution: Setup

Variable	x (horizontal)	y (vertical)
Displacement	$\Delta x = ?$	$\Delta y = -78.4 \text{ m}$
Initial Velocity	$v_{x,i} = 15.0 \text{ m/s}$	$v_{y,i} = 0 \text{ m/s}$
Final Velocity	$v_{x,f} = 15.0 \text{ m/s}$	?
Acceleration	$a_x = 0$	$a_y = -9.80 \text{ m/s}^2$
Time	$t = ?$	$t = ?$

Strategy: Solve for time using vertical motion, then use that time to find horizontal distance.

Example Solution: Part A

A) How long does it take to hit the ground?

- Use the vertical motion equation:

$$\Delta \vec{y} = \vec{v}_{y,i}t + \frac{1}{2} \vec{a}_g t^2$$

- Since $v_{y,i} = 0$, the equation simplifies to:

$$\Delta \vec{y} = \frac{1}{2} \vec{a}_g t^2$$

Example Solution: Part A (continued)

- Rearrange to solve for t :

$$t = \sqrt{\frac{2\Delta \vec{y}}{\vec{a}_g}}$$

- Now substitute values:

$$t = \sqrt{\frac{2(-78.4 \text{ m})}{-9.80 \text{ m/s}^2}} = \sqrt{16.0 \text{ s}^2} = 4.00 \text{ s}$$

- **Note:** Mathematically, $t = -4.00 \text{ s}$ is also a solution to $t^2 = 16.0 \text{ s}^2$, but we reject it because negative time has no physical meaning in this context.

Example Solution: Part B

B) How far from the base does it land?

- Use the horizontal motion equation:

$$\Delta \vec{x} = \vec{v}_x t$$

- Substitute values:

$$\Delta \vec{x} = (15.0 \text{ m/s})(4.00 \text{ s}) = 60.0 \text{ m}$$

- **Answer:** The stone lands **60.0 m** from the base of the cliff.

Check for Understanding

A cannonball is fired horizontally from a cannon. At the exact same instant, another identical cannonball is simply dropped from the same height.

Which one hits the ground first? (Ignore air resistance.)

- **Answer:**
They hit the ground at the exact same time.
- Their horizontal motion has no effect on their vertical motion.
- They both start with $v_{y,i} = 0$ and accelerate downwards at a_g .

Answering Our Essential Questions

1. How can a complex, two-dimensional motion be broken down into simpler parts?

By resolving it into two independent, perpendicular components: horizontal (x) and vertical (y).

2. Do the vertical and horizontal motions of a projectile affect each other?

No. Gravity only acts vertically, so horizontal velocity is constant. The horizontal motion does not change the way the object falls.

3. How can we use our 1D kinematic equations to solve 2D problems?

Apply the constant velocity equation for the horizontal motion ($\Delta x = v_x t$).

Apply the constant acceleration equations for the vertical motion.

Use time (t) as the variable that connects the two analyses.

Summary and Next Steps

The Core Idea:

Analyze 2D projectile motion as two separate 1D problems.

- Horizontal: Constant velocity ($a_x = 0$)
- Vertical: Constant acceleration ($\vec{a}_g$)
- Time (t) is the link between them.

Bridge Forward:

In Problem Set 3.1, you will apply this technique to solve a variety of problems involving objects launched horizontally. This is the foundational skill for the entire unit.